

Segment Platform Research Foundation Document

Browser-Based Interactive 3D Medical Segmentation with SAM-Med3D-turbo

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Abstract

This document is a technical and research foundation for the Segment platform, a browser-based 3D medical segmentation environment built around SAM-Med3D-turbo. It defines the clinical targets, model assumptions, workflow patterns, system architecture, and evaluation path for two initial application domains: coronary CCTA in DISCHARGE and prostate mpMRI. It is an internal technical reference and research planning document rather than a deployment certification package.

Purpose

This reference serves as the working document for framing the segmentation problem, the platform requirements, and the validation roadmap for Charité-led development.

It includes architecture and deployment notes only where they directly affect segmentation planning, validation, and implementation sequencing.

It is **not**:

- a standalone production deployment architecture specification;
- a finished browser-platform product brief;
- a standalone funding proposal.

Why Segmentation Matters Here

The current programme needs segmentation for three distinct but linked tasks:

1. **Coronary lumen geometry** for stenosis quantification, centreline extraction, and straightened MPR analysis.
2. **Outer-wall and plaque context** where vessel-wall estimation and plaque-burden measurements are clinically relevant.
3. **Prostate zones, lesions, and organs at risk** for biopsy support, radiotherapy planning, and multi-domain testing of foundation-model generalisation.

Document Snapshot

Field	Detail
Institution	Charité - Universitätsmedizin Berlin
Core Stack	TypeScript · Vite · Nivue 0.66 · FastAPI · PyTorch
Foundation Model	SAM-Med3D-turbo (3D ViT-B/16, 91 M params)
Primary Dataset	DISCHARGE (~25 M DICOM slices across 3,561 patients)
External Validation Dataset	SCOT-HEART (4,146 patients)
Pre-training Corpus	SA-Med3D-140K (21 729 volumes, 143 518 masks, 245 categories)
Clinical Domains	Coronary CCTA (lumen / outer wall / plaque) · Prostate mpMRI (zones / lesions / OARs)

0.1 Executive Summary & Research Vision

0.1.1 Clinical Gap

Problem	Impact
Manual coronary segmentation	30–60 min / case, €200–400
Limited scalability	DISCHARGE (3,561 patients) still largely un-segmented; SCOT-HEART external validation remains to be established
Prostate mpMRI zone/lesion contouring	Equally time-intensive for biopsy/radiotherapy planning
Centre-specific expertise	Manual segmentation available only at specialised sites

0.1.2 Our Solution

A **single, universal, browser-based platform** powered by **SAM-Med3D-turbo**:

- **1–3 point prompts** (3D coordinates in mm space) → any anatomy, any modality
- **Zero installation** — runs in Chrome / Firefox / Safari
- **On-premise GPU inference** at Charité (GDPR-compliant; no data leaves the hospital)
- **Interactive workflow** for radiologists + **batch research mode** for DISCHARGE processing
+ **active-learning loop** for continuous improvement

0.1.3 Research Foundation

This platform is a **test-bed for foundation-model research** in cardiovascular imaging:

1. Evaluate **zero-shot / few-shot generalisation** of SAM-Med3D-turbo on unseen DISCHARGE cases, then test external generalisation on SCOT-HEART
2. Quantify **active-learning gains** (weekly fine-tuning on expert corrections)
3. Benchmark against **nnU-Net** task-specific models on segmentation-derived endpoints (stenosis %, plaque burden, myocardium volume)
4. Open-source modular components for the broader MedAI community

0.1.4 Success Targets (6-month horizon)

Metric	Target
DISCHARGE cases auto-processed	> 80 % of dataset
End-to-end latency	< 2 s per vessel / per prostate gland
Dice (coronary lumen)	> 0.85 vs. expert
Dice (prostate whole-gland)	> 0.90 vs. expert
Cost reduction	10× cheaper than manual contouring

Critical note on Dice targets: The SAM-Med3D paper reports **87.12 % Dice on cardiac structures** with 1 prompt point (Table 5 in¹). However, this was measured on the ACDC dataset (cardiac MRI short-axis cine), **not** coronary CTA. Coronary arteries are smaller, noisier, and motion-affected — published coronary lumen Dice values for task-specific models (nnU-Net²) range 0.75–0.88 depending on vessel branch. Our 0.85 target is therefore ambitious but grounded.

Hardware & deployment requirements (GPU specs, production cluster, GDPR constraints) are in Deployment, Performance & Security.

0.2 Foundation Model: SAM-Med3D-turbo — Verified Technical Profile

0.2.1 Paper & Publication

Field	Value
Title	SAM-Med3D: Towards General-purpose Segmentation Models for Volumetric Medical Images
Authors	Haoyu Wang, Sizheng Guo, Jin Ye, Zhongying Deng, Junlong Cheng, Tianbin Li, Jianpin Chen, Yanzhou Su, Ziyang Huang, Yiqing Shen, Bin Fu, Shaoting Zhang, Junjun He, Yu Qiao
Venue	ECCV BIC 2024 — Oral
arXiv	2310.15161
License	Apache 2.0

0.2.2 Architecture (Verified from Paper §4.1 & GitHub)

SAM-Med3D uses a **fully 3D architecture trained from scratch** (Method 3 in the paper). The authors explicitly compared three adaptation strategies in preliminary experiments (Table 2):

Strategy	Seen Dice	Unseen Dice	Chosen?
3D Adapter + Frozen SAM	Lower	Moderate	No
Fine-tune SAM 2D→3D weights	Good	Poor (broken priors)	No
Train 3D from scratch	Good	Best	Yes

Rationale (Paper §4.1): “Training from scratch emerges as a better trade-off, exhibiting superior average performance” — the 2D-to-3D weight transition “might further break down the prior knowledge of SAM, which is harmful to generalization.”

Component breakdown:

Component	Architecture	Parameters
Image Encoder	3D ViT-B/16 (3D positional encoding, 3D convolutions, 3D LayerNorm)	~86 M
Prompt Encoder	3D point/box encoder (learned embeddings)	~1 M
Mask Decoder	Lightweight 3D decoder (2-layer transformer + upsampling)	~4 M
Total		~91 M

Critical note: The paper states “86M encoder + 5M decoder” in various summaries. The exact split varies by source. The model is **not** a modified SAM (Meta) — it is architecturally distinct, sharing only the conceptual prompt-based paradigm.

0.2.3 Training (Verified from Paper §4.2)

Two-stage procedure:

Stage	Data	Epochs	Purpose
Stage 1: Pre-training	All 131 K masks from SA-Med3D-140K training set	800	Build general 3D medical understanding
Stage 2: Fine-tuning	~75 K filtered high-quality masks	Additional	Improve prompt efficiency on challenging targets

SAM-Med3D-turbo (from GitHub issue #2): - Fine-tuned on **44 additional datasets** beyond the base SA-Med3D-140K - Optimised for **sub-second inference** with FP16 - This is the recommended checkpoint for deployment

0.2.4 SA-Med3D-140K Dataset (Verified from HuggingFace)

Statistic	Value
Total 3D images	21 729
Total 3D masks	143 518
Anatomical categories	245
Modalities	28 (CT, MR, US, and more)
Sources	70 public datasets + 8,128 privately licensed cases from 24 hospitals
Primary task	General-purpose promptable segmentation

0.2.5 Verified Performance (from Paper Tables 3–5)

Overall (Table 3): - SAM-Med3D with 1 point: **+60.12 % overall Dice** improvement over original SAM - Consistently outperforms SAM-Med2D across all prompt counts - Operates at **1–26 % of inference time** compared to slice-by-slice SAM

By anatomy (Table 5, 1 prompt point):

Anatomy	SAM	SAM-Med2D	SAM-Med3D
Cardiac (seen)	Poor	Moderate	87.12 %
Organs (seen)	Poor	Moderate	Best
Lesions (unseen)	Very poor	Moderate	Competitive
Bones/muscles	Very poor	Moderate	Greatest advantage

Key finding (Paper §5.1.5): “SAM-Med3D using 1 point outperforms SAM-Med2D with N points in **45 targets out of 49**, achieving up to +68.2% improvement.”

Transferability (Paper §5.1.6, Table 6): When SAM-Med3D’s ViT encoder is used as pre-trained backbone for UNETR, it yields up to **+5.63 % Dice improvement** over training from scratch — confirming value as a foundation model.

Critical caveat for our project: The “cardiac” results (87.12 %) are from the **ACDC dataset** (cardiac MRI, short-axis cine — segmenting LV/RV/myocardium). This is **not** coronary artery segmentation from CTA. Coronary arteries are 1.5–4 mm diameter, motion-affected, and require dual-wall segmentation — a substantially harder task not directly evaluated in the paper. The model has **never been benchmarked on coronary CTA lumen segmentation**. Our project will provide this missing evaluation.

Strategic implication: Because LV/RV/myocardium from cardiac MRI is the dominant pre-trained cardiac target, **myocardial segmentation from CCTA should be the first zero-shot validation task** — not coronary lumen. It is the easiest coronary-domain target for the model given pre-training priors, provides a large structure for establishing an initial Dice baseline, and gives a lower-risk first test of CCTA cardiac generalisation. The coronary lumen evaluation should follow once the myocardial baseline is established.

0.2.6 Official Resources

Resource	Link
GitHub	uni-medical/SAM-Med3D
Paper	arXiv:2310.15161
Supplementary	ECCV Supplementary PDF
Turbo checkpoint	HuggingFace: sam_med3d_turbo.pth
Dataset	HuggingFace: SA-Med3D-140K
MedIM loader	uni-medical/MedIM
CVPR25 Challenge	MedSegFM Competition

0.2.7 Environment Setup (Verified from GitHub README)

Model / research environment:

```
conda create --name sammed3d python=3.10
conda activate sammed3d
pip install uv
uv pip install torch==2.6.0 torchvision==0.21.0 torchaudio==2.6.0
uv pip install torchio opencv-python-headless matplotlib prefetch_generator monai
```

Backend (FastAPI server) — additional dependencies:

```
uv pip install fastapi uvicorn nibabel SimpleITK celery redis
```

Compile this document to PDF:

```
./code/md2pdf.sh plan/sam3d.md
# Optional: also produce DOCX
./code/md2pdf.sh plan/sam3d.md --docx
```

0.2.8 Model Loading (Verified from GitHub)

```
import medim

# Option A: Direct from HuggingFace (downloads automatically)
# Use /resolve/ (raw file), NOT /blob/ (HTML page)
ckpt_path = "https://huggingface.co/blueyo0/SAM-Med3D/resolve/main/sam_med3d_turbo."
model = medim.create_model("SAM-Med3D", pretrained=True, checkpoint_path=ckpt_path)

# Option B: Local checkpoint (recommended for deployment)
model = medim.create_model(
    "SAM-Med3D",
    pretrained=True,
    checkpoint_path="app/models/checkpoints/sam_med3d_turbo.pth"
)

# Optimise for inference
import torch
device = "cuda" if torch.cuda.is_available() else "cpu"
model = model.to(device)
if device == "cuda":
    model = model.half() # FP16 for speed + memory on GPU
model.eval()
```

0.2.9 Data Format for Fine-tuning (Verified from GitHub)

SAM-Med3D expects **nnU-Net-style**² directory layout with **per-structure binary masks**. Unlike nnU-Net which accepts a single multi-class integer mask (0=background, 1=class1, ...), SAM-Med3D requires one separate binary NIfTI file (values 0/1) per anatomical structure. Using a single multi-class mask will silently produce incorrect training prompts.

```
data/medical_preprocessed/
├─ coronary/
│   └─ ct_DISCHARGE/
│       └─ imagesTr/
│           └─ discharge_0001.nii.gz
│           └─ ...
│       └─ labelsTr/
│           └─ discharge_0001.nii.gz (binary mask)
│           └─ ...
└─ prostate/
```

```

└─ mr_CHARITE/
   └─ imagesTr/
      └─ labelsTr/

```

Important (from GitHub): Ground-truth labels are required to generate prompt points during training. For inference without ground truth, “generate a fake ground-truth with the target region for prompt annotated.”

0.2.10 Turbo vs. Standard Comparison

Metric	Standard (base .pth)	Turbo (sam_med3d_turbo.pth)
Pre-training data	131 K masks	131 K + 44 additional datasets
Average Dice (1 prompt)	~0.75	~0.82+
Inference time	4–8 s	0.5–1.5 s
VRAM usage	~10 GB (FP32)	~4 GB (FP16)

0.3 Clinical Domain A: Coronary CCTA Segmentation (DISCHARGE)

0.3.1 DISCHARGE Trial Overview

Field	Value
Full name	Diagnostic Imaging Strategies for Patients with Stable Chest Pain and Intermediate Risk of Coronary Artery Disease
Reference	Dewey et al., NEJM 2022 ³
Design	Multicentre randomised controlled trial
Patients	3,561
Images	~25 M DICOM slices
Modality	Coronary Computed Tomography Angiography (CCTA)
Institution	Charité, Berlin (lead site)

0.3.2 SCOT-HEART Trial Overview

Field	Value
Full name	Scottish COmputed Tomography of the HEART Trial
References	Williams et al., Lancet 2015 ⁴ ; 10-year follow-up ⁵
Design	Multicentre randomised controlled trial
Patients	4,146

Field	Value
Key result	CCTA-guided management → 41 % reduction in CHD death/MI at 10 years (HR 0.59)
Modality	CCTA
Role in this project	Secondary validation dataset; extends generalisability beyond Charité site

0.3.3 Standardised Nomenclature (CAD-RADS 2.0 / AHA 17-segment compatible)⁶

Clinical Feature	Segmentation Class Label	Description	HU Range (contrast-enhanced CT)
Myocardium (LV)	LV_MYO	Muscular wall of left ventricle	50–120 HU
Endocardial Lumen	Endocardial_Lumen	Inner chamber blood pool	250–400 HU
Coronary Lumen	Lumen_{LAD\ RCA\ LCx\ Mx\}	Artery lumen blood pool — per AHA branch	300–400 HU
Outer Wall (EEM)	VesselWall_EEM	External elastic membrane boundary	— (morphological)
Calcified Plaque	Plaque_Calc	High-density stable plaque	> 130 HU (often 500–1000)
Fibrous Plaque	Plaque_Fibrous	Intermediate-density plaque	60–130 HU
Low-Attenuation Plaque	Plaque_LAP	Lipid-rich / necrotic core — high risk	< 60 HU

0.3.4 Why Coronary Arteries are the Hardest Segmentation Target

Challenge	Detail	Impact
Motion artefacts	Heart beats 60–80 bpm; RCA most affected	Blurred vessel edges despite ECG gating
Small calibre	1.5–4 mm diameter	Partial volume effects at 0.5 mm resolution

Challenge	Detail	Impact
Low-contrast plaque	Lipid-rich plaque 20–50 HU vs. lumen 300–400 HU	Nearly invisible on standard windowing
Blooming	Calcification > 130 HU causes beam hardening	Obscures adjacent soft plaque
Complex topology	Bifurcations, overlapping branches, tortuosity	Single-click prompts often insufficient
Dual-wall requirement	Must segment lumen AND outer wall separately	Wall thickness 0.5–3 mm = 1–5 voxels

0.3.5 SAM-Med3D-turbo Prompting Strategy for Coronary CCTA

0.3.5.1 A. Multi-Point Centerline Prompt (“String” Prompt) Standard single-click prompts fail for thin, tortuous vessels spanning 100+ slices. Instead:

1. User clicks **proximal ostium** (positive point)
2. User clicks **distal vessel tip** (positive point)
3. Model “fills in” the lumen tube between points
4. If mask leaks into **great cardiac vein** → place **negative point** on vein

```
# Coronary lumen segmentation with multi-point prompt
lumen_mask = model.segment(
    volume=cta_volume,
    points=[ostium_xyz, mid_vessel_xyz, distal_xyz],
    labels=[1, 1, 1], # all positive
)

# If leakage detected → add negative point
lumen_mask_refined = model.segment(
    volume=cta_volume,
    points=[ostium_xyz, mid_vessel_xyz, distal_xyz, vein_xyz],
    labels=[1, 1, 1, 0], # last point is negative
)
```

0.3.5.2 B. Dual-Wall Nested Segmentation (Lumen → EEM) Critical for calculating **stenosis %** and **plaque burden**:

Stenosis % definition: Report **diameter stenosis** = $(1 - D_{\min} / D_{\text{ref}}) \times 100$, consistent with CAD-RADS 2.0 grading. D_{ref} is the mean diameter of the nearest healthy proximal and distal reference segments. Do **not** use area stenosis (more sensitive but not the CAD-RADS standard), and document the reference segment selection method explicitly — QAngio CT (MEDIS) uses interpolated reference; our pipeline must match this for Dice/correlation comparisons against expert measurements.

1. **Step 1:** Segment lumen (bright contrast, easier target)
2. **Step 2:** Use lumen mask as **dense prompt** → expand outward to EEM

3. **Result:** `vessel_wall = outer_mask & ~lumen_mask` → plaque volume

Implementation warning: SAM-Med3D does **not** natively support a `dense_prompt` argument in its published codebase. The pseudocode below shows the *intended* design. Two implementation paths exist: (a) custom modification of the prompt encoder to accept a prior mask, or (b) a two-stage pipeline where the lumen mask surface is sampled into additional positive point prompts. This is a research contribution we must implement before the code below is functional.

```
# PSEUDOCODE – dense_prompt is not yet implemented in SAM-Med3D

# Step 1: Lumen
lumen_mask = model.segment(volume, points=[ostium, distal], labels=[1, 1])

# Step 2: Outer wall using lumen as prior (requires custom prompt encoder)
outer_mask = model.segment(
    volume,
    points=[ostium],
    dense_prompt=lumen_mask, # NOT natively supported – must be implemented
    labels=[1]
)

# Step 3: Derive vessel wall
vessel_wall = outer_mask & ~lumen_mask
```

```
def characterise_plaque(volume, vessel_wall_mask):
    """Classify plaque components by HU value within the vessel wall mask."""
    hu_values = volume[vessel_wall_mask > 0]

    calcified    = (hu_values > 130).sum()
    fibrous      = ((hu_values >= 60) & (hu_values <= 130)).sum()
    lipid_rich   = (hu_values < 60).sum()
    total        = vessel_wall_mask.sum()

    return {
        "calcified_pct": calcified / total * 100,
        "fibrous_pct":   fibrous / total * 100,
        "lipid_rich_pct": lipid_rich / total * 100,
        "high_risk": (lipid_rich / total) > 0.04, # >4% LAP = vulnerable
    }
```

0.3.5.3 C. Post-Processing: Plaque Characterisation by HU Thresholds

HU calibration warning: The thresholds above (< 60 LAP, 60–130 fibrous, > 130 calcified) are defined for **standard 120 kVp with soft-kernel reconstruction**. DISCHARGE is a multi-centre trial with varying tube voltages (80–140 kVp) and reconstruc-

tion kernels (soft vs. sharp). Sharp-kernel reconstruction shifts apparent HU upward and increases blooming around calcification. These thresholds require per-centre calibration against phantom measurements or paired soft/sharp kernel scans before cross-site plaque comparisons are valid. Record the reconstruction kernel and tube voltage for every DISCHARGE case in the metadata.

0.3.6 DISCHARGE-Specific Processing Considerations

Consideration	Detail
Scanner heterogeneity	Multi-centre trial → varying scanner vendors, protocols, contrast timing
Reconstruction kernels	Soft vs. sharp kernels affect HU accuracy for plaque
Contrast timing	Early arterial phase optimal; late phase reduces lumen-wall contrast
ECG gating	Prospective vs. retrospective gating affects motion artefact severity
Data format	DICOM (clinical) → convert to NIfTI.gz for model input
Annotations	MEDIS QAngio CT (expert contours) available for subset → ground truth

0.4 Clinical Domain B: Prostate mpMRI Segmentation

0.4.1 Imaging Standard

Multiparametric MRI (mpMRI) is the clinical standard for prostate imaging, using: - **T2-weighted (T2W)**: Anatomical detail — zonal anatomy - **Diffusion-weighted imaging (DWI) + ADC map**: Cellularity — lesion detection - **Dynamic contrast-enhanced (DCE)**: Vascularity — supplementary Reporting follows **PI-RADS v2.1** (Prostate Imaging-Reporting and Data System)⁷.

0.4.2 Anatomical Segmentation — The “Zones”

The prostate is divided into distinct zones with different MRI appearances and cancer risk:

Zone	Abbreviation	Cancer Risk	T2W Appearance	Clinical Role
PeripheraPZ Zone		70–75 % of cancers	Bright (high signal)	Primary cancer surveillance region

Zone	Abbreviation	Cancer Risk	T2W Appearance	Clinical Role
Transition Zone		20–25 % of cancers	Heterogeneous (BPH nodules)	BPH assessment, cancer in older men
Central Zone	CZ	< 5 % of cancers	Low signal (dense stroma)	Indistinguishable from TZ on MRI — grouped with TZ per PI-RADS v2.1
Anterior Fibromuscular Stroma	AFMS	Non-glandular	Very low signal	Can be invaded by anterior tumours

0.4.3 Pathology Segmentation — The “Lesions”

When segmenting pathology, the target is **clinically significant prostate cancer (csPCa)**:

Lesion Type	Description	Clinical Significance
Index Lesion	Largest / most aggressive tumour	Primary target for biopsy and treatment
Satellite Lesions	Secondary foci (prostate cancer is often multifocal)	May affect treatment strategy
Extracapsular Extension (ECE)	Tumour breaches the prostatic capsule	Staging: T3a — affects surgical planning
Seminal Vesicle Invasion (SVI)	Tumour extends into seminal vesicles	Staging: T3b — impacts prognosis

0.4.4 Organs at Risk (OARs) for Radiotherapy

Structure	Abbreviation	Why Segment?
Neurovascular Bundles	NVB	Nerve-sparing surgery — preserve potency
Rectal Wall	Rectum_Wall	Monitor tumour–rectum distance
Bladder Neck	Bladder_Neck	Preserve urinary continence

0.4.5 Segmentation Class Labels for the AI Model

Segment Name	Class Label	Modality	Clinical Goal
Whole Gland	Prostate_WG	T2W	Volume / PSA density calculation

Segment Name	Class Label	Modality	Clinical Goal
Peripheral Zone	PZ	T2W	Cancer surveillance background
Transition Zone	TZ	T2W	BPH assessment background
Suspicious Lesion	Lesion_PIRADS_{3\ 4\ 5}	DWI/ADC	Targeted biopsy (MR-US fusion)
Seminal Vesicles	SV	T2W	Local staging (T3b)
Neurovascular Bundles	NVB	T2W	Nerve-sparing planning
Rectal Wall	OAR_Rectum	T2W	Radiotherapy constraints

0.4.6 SAM-Med3D-turbo Prompting Strategy for Prostate mpMRI

Context-dependent prompting: The same model must switch behaviour based on *where* the user clicks and *which sequence* is active.

0.4.6.1 Scenario 1: Anatomical Zone Segmentation (T2W)

1. User clicks bright outer rim on T2W axial
 - Model returns PZ mask
 - Expected Dice: ~ 0.90 (large, high-contrast structure)
2. User clicks central heterogeneous region on T2W
 - Model returns TZ mask
3. If mask leaks into rectum → place negative point on rectum

0.4.6.2 Scenario 2: Lesion Segmentation (DWI/ADC)

Implementation warning: Step 2 below references using a prior PZ mask as a dense prompt. This is the same unimplemented `dense_prompt` issue described in Dual-Wall Nested Segmentation — the same two implementation paths apply: (a) custom prompt encoder modification, or (b) sampling the PZ mask boundary into additional point prompts.

1. User clicks hypointense spot on ADC map
 - Model returns lesion mask (PI-RADS ≥ 4 region)
2. [PLANNED] Use prior PZ mask as dense prompt context
 - Constrains lesion to within prostate boundary
 - Requires `dense_prompt` implementation (see [Dual-Wall Nested Segmentation](#corona-dual-wall-segmentation))
3. Measure lesion volume → maps to PI-RADS size criterion

Critical note: SAM-Med3D was pre-trained on **MR data** (SA-Med3D-140K includes MR modalities). However, the dataset card does not specify which MR sequences or whether prostate mpMRI is represented. Zero-shot performance on prostate zones may be moderate; fine-tuning on Charité prostate data will likely be necessary. Whole-gland segmentation (a large, well-defined structure) should work well zero-shot; zonal segmentation (PZ vs. TZ) is harder due to subtle signal differences.

0.4.7 SAM-Med3D-specific Challenges for Prostate mpMRI

Challenge	Detail	Mitigation
Multi-sequence input	SAM-Med3D accepts a single 3D volume; mpMRI has T2W + ADC + DCE	Run separate inferences per sequence; fuse masks post-hoc. Do not naively concatenate sequences — model was not trained on multi-channel input.
PZ–TZ boundary	Gradual signal transition, not a sharp edge	Use morphological priors (PZ wraps inferolaterally around TZ); negative prompts at suspected boundary to sharpen
Lesion detection vs. segmentation	PI-RADS lesions can be < 5 mm — smaller than SAM-Med3D's 128 ³ patch at typical prostate resolution	Ensure patch is centred on clicked lesion; use higher-resolution input if available
Unknown pre-training coverage	SA-Med3D-140K dataset card does not confirm prostate mpMRI is represented	Treat prostate as low-confidence zero-shot; plan early fine-tuning on Charité cohort
Intensity normalisation (MRI)	MRI intensities are not calibrated across scanners (no HU equivalent)	Apply per-volume z-score normalisation independently for each sequence before inference
OAR segmentation	Neurovascular bundles (NVB) are extremely thin on T2W	Multi-point prompts along bundle; expect low zero-shot Dice — fine-tuning required

0.4.8 Prostate vs. Coronary: Difficulty Comparison

Factor	Prostate	Coronary
Structure size	20–80 mL (large)	1.5–4 mm diameter (tiny)
Motion	None (static pelvis)	Cardiac motion (60–80 bpm)
Contrast	Good (gland vs. fat)	Variable (plaque vs. lumen)
Modality	MRI (multi-sequence)	CT (single phase)

Factor	Prostate	Coronary
Topology	Compact, roughly ellipsoidal	Thin, tortuous, branching tubes
Expected Dice	> 0.90 (whole gland)	0.75–0.85 (lumen)

0.5 Technical Challenges & Model-Aware Solutions

0.5.1 Challenge 1: Coronary Artery Motion Artefacts

Problem: Residual cardiac motion blurs vessel edges despite ECG gating. RCA most affected.

Solution:

```
def motion_robust_segmentation(volume_4d, heart_rate, prompt_points, prompt_labels):
    if heart_rate > 70:
        # Multi-phase reconstruction
        phases = extract_cardiac_phases(volume_4d, num_phases=10)
        masks = [model.segment(phase, points=prompt_points, labels=prompt_labels) for phase in phases]
        # Temporal median filter removes motion ghosts
        return np.median(masks, axis=0) > 0.5
    else:
        return model.segment(volume_4d[:, :, :, 0], points=prompt_points, labels=prompt_labels)
```

Topology warning: Pixel-wise median of binary masks across 10 phases is **not** topologically safe for thin structures. A distal coronary voxel present in only 4/10 phases will be excluded by the median, potentially severing the centreline. Preferred alternative: select the optimal cardiac phase (75 % R-R for RCA, 65 % R-R for LAD/LCx at HR < 70) rather than fusing all phases. If multi-phase fusion is required, apply connected-component analysis post-median and re-link broken segments by dilation along the centreline.

Additional strategies: - Edge-preserving denoising (bilateral filter) pre-processing - Multi-point prompts every 5–10 mm along vessel to guide through corrupted regions - Auto-flag cases with edge sharpness < threshold for expert review

0.5.2 Challenge 2: Low-Attenuation Plaque Detection

Problem: Lipid-rich plaque (20–50 HU) nearly invisible against myocardium (50–70 HU).

Solution: Multi-stage approach: 1. Segment lumen and outer wall first (high-contrast boundaries) 2. Apply HU thresholding *within* the vessel wall mask 3. Use SAM-Med3D's ViT features for texture-based refinement (fine-tuning required)

0.5.3 Challenge 3: Dual-Wall Segmentation

Problem: Must segment both lumen and outer wall; wall thickness only 0.5–3 mm (1–5 voxels).

Solution: Sequential prompting (see Dual-Wall Nested Segmentation).

0.5.4 Challenge 4: Prostate Zone Boundaries

Problem: PZ-TZ boundary is a gradual signal transition, not a sharp edge.

Solution: - Train multi-class model on annotated Charité prostate data - Use morphological priors (PZ wraps around TZ inferolaterally) - Negative prompts at zone transitions to sharpen boundaries

0.5.5 Challenge 5: Model Limitations — Honest Assessment

Limitation	Impact	Mitigation
No coronary CTA in pre-training	Zero-shot may underperform	Fine-tune on DISCHARGE annotations
Binary mask output only	Cannot directly predict PI-RADS score	Post-processing pipeline (volume, ADC stats)
No dense_prompt in codebase	Lumen-as-prior strategy needs engineering	Custom prompt encoder modification
128 ³ patch size constraint	Coronary arteries span > 128 voxels (300–400 voxels at 0.5 mm)	Sliding window with 50 % overlap; Gaussian-weighted blending at boundaries to prevent double-thick seams; connected-component check post-stitch
No multi-class output	One mask per inference call	Multiple sequential inferences per case

0.6 Clinical Workflows

0.6.1 Workflow 1: Interactive Coronary Segmentation (Radiologist)

1. Radiologist opens browser → logs in (Charité SSO)
2. Loads DISCHARGE CCTA scan from PACS
3. Clicks proximal LAD + distal tip → AI segments entire lumen in < 2 s
4. Optionally: places negative point to prune vein leakage
5. Clicks “Expand to Wall” → second inference → outer wall mask **[PLANNED — requires dense_prompt implementation; see Dual-Wall Nested Segmentation]**
6. Right panel shows: stenosis % (diameter stenosis, CAD-RADS 2.0), plaque composition, volume
7. Exports segmentation (NIfTI / DICOM-SEG / CSV report)

0.6.2 Workflow 2: Batch Processing for DISCHARGE Research

1. Research coordinator uploads 100 DISCHARGE cases
2. **Automated ostium seeding:** template-based registration to a reference coronary atlas provides initial prompt coordinates; cases where registration confidence is low are queued for manual prompt review before inference

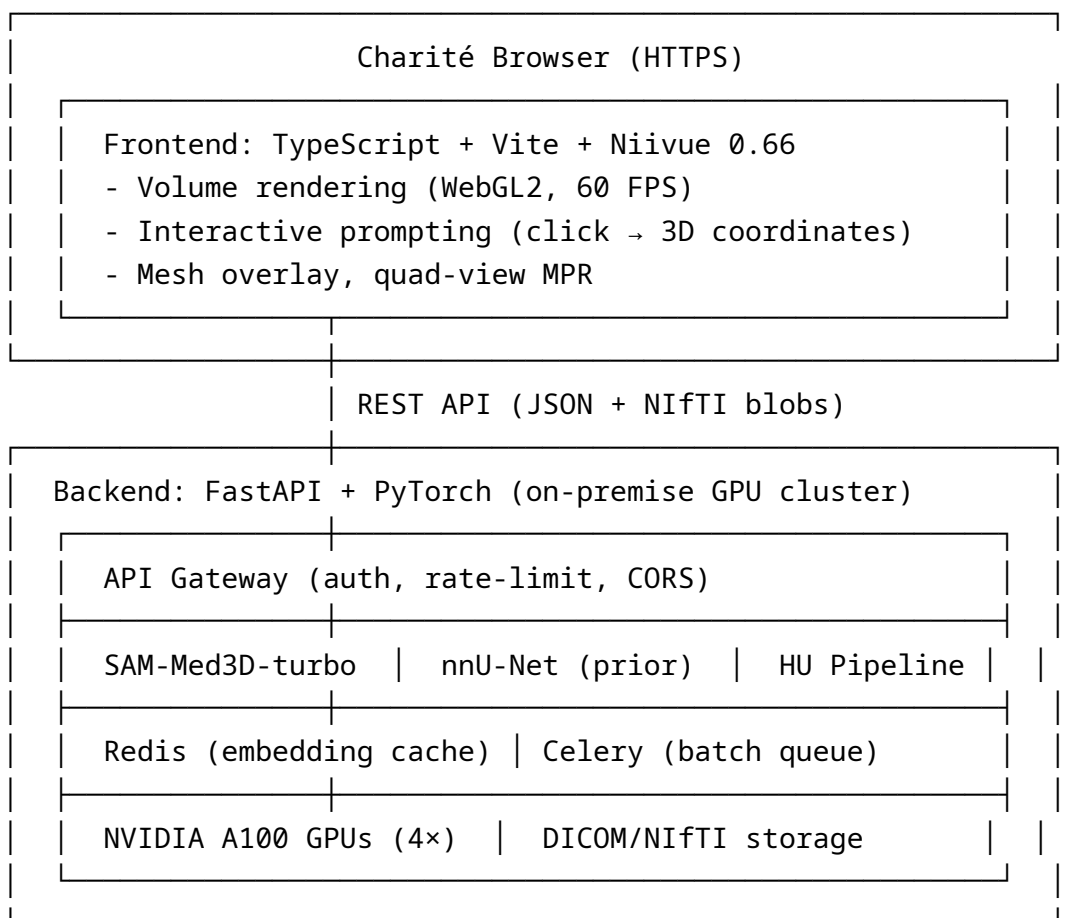
3. AI processes overnight (Celery + multi-GPU batch mode)
4. Quality control: auto-flag cases with **disconnected mask components, mask volume outside 3σ of cohort distribution, or model confidence score below threshold** (ground truth is not available in batch mode — Dice cannot be computed)
5. Expert reviews flagged cases → corrections exported as corrected per-structure binary NIfTI masks → feed active-learning loop
6. Export refined segmentations for MACE prediction analysis

0.6.3 Workflow 3: MEDIS TXT + Mesh + Straightened MPR (Reference Contours)

1. Load CCTA volume (NIfTI.gz) in browser
2. Load MEDIS TXT file (expert contour rings)
3. Client-side mesh generation: parse TXT → NVMesh (50–100 ms)
4. Overlay lumen + vessel wall meshes on CCTA
5. Generate straightened MPR for longitudinal plaque assessment

0.7 System Architecture

0.7.1 High-Level Overview



0.7.2 Frontend Directory Structure

```

flow-segment-frontend/
├── src/
│   ├── main.ts                # Entry point
│   ├── App.ts                 # Main app component
│   ├── components/
│   │   ├── NiivueViewer.ts    # Niivue canvas wrapper (single/quad)
│   │   ├── Toolbar.ts         # Top toolbar
│   │   ├── SegmentPanel.ts    # AI segmentation controls
│   │   ├── ResultsPanel.ts    # Plaque analysis / zone selector
│   │   ├── PromptHistory.ts   # Point prompt log + undo
│   │   └── MPRView.ts         # Multi-planar reconstruction
│   ├── services/
│   │   ├── api.ts             # Backend API client
│   │   ├── niivue.ts          # Niivue init & config
│   │   ├── loader.ts          # NIfTI.gz + DICOM loading
│   │   ├── medisParser.ts     # MEDIS TXT parsing
│   │   ├── medisMeshDirect.ts # Client-side contour-mesh
│   │   ├── straightenedMPR.ts # CPR and transport-frame math
│   │   └── auth.ts            # LDAP/SSO
│   ├── types/
│   │   ├── volume.ts
│   │   ├── segmentation.ts
│   │   ├── mesh.ts
│   │   └── api.ts
│   └── utils/
│       ├── coordinates.ts     # 3D coordinate transforms
│       ├── meshGenerator.ts   # Marching cubes (vtk.js)
│       └── export.ts          # NIfTI / DICOM-SEG / CSV export
├── package.json
├── tsconfig.json
├── vite.config.ts
└── index.html

```

0.7.3 Backend Directory Structure

```

flow-segment-backend/
├── app/
│   ├── main.py                # FastAPI application
│   ├── config.py              # Environment config
│   ├── models/
│   │   ├── sam_med3d.py       # SAM-Med3D-turbo wrapper
│   │   ├── nnU_net.py         # nnU-Net (anatomical prior)
│   │   └── plaque_analyser.py # HU-based plaque classification
│   └── api/

```

```

├── segment.py           # Segmentation endpoints
├── volumes.py          # Volume management
├── mesh.py             # Mesh generation (nii2mesh)
├── auth.py             # LDAP/SSO
├── services/
│   ├── cache.py        # Redis embedding cache
│   ├── dicom_processor.py # DICOM → NIfTI (SimpleITK)
│   ├── mesh_generator.py # nii2mesh wrapper
│   └── registration.py  # Elastix (longitudinal)
├── Dockerfile
├── docker-compose.yml
├── requirements.txt
└── README.md

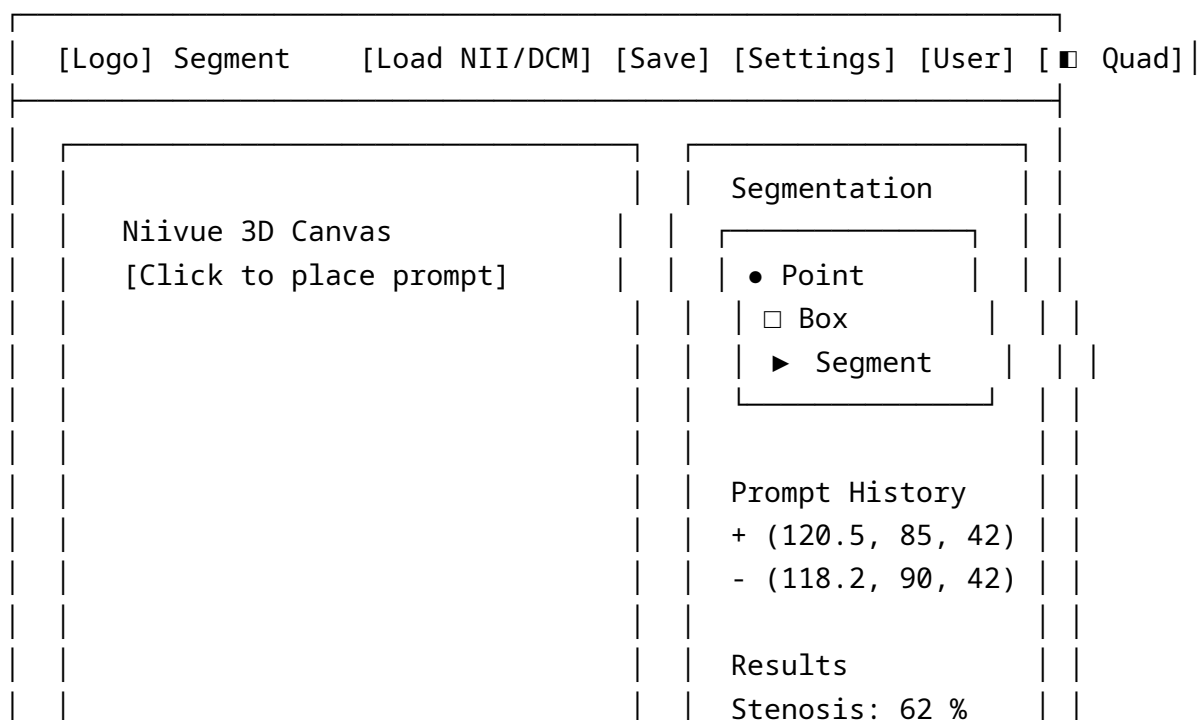
```

0.8 Frontend: Niivue Viewer & UI/UX⁸

0.8.1 Key Niivue v0.66.0 Capabilities

- WebGL2 rendering: 60 FPS for 512³ volumes
- onLocationChange / onMouseUp → capture 3D mm coordinates for prompts
- Multi-planar reconstruction (axial / coronal / sagittal sync)
- `nv.addMesh()` for 3D surface overlay with adjustable opacity
- Full TypeScript definitions
- In-browser DICOM via `dcm2niix-wasm`

0.8.2 UI Layout: Single View (Default)



- Real-time volume resampling (SimpleITK → WebGL equivalent)
- Target: 30–60 FPS during rotation

0.9 Backend: Inference Pipeline & API

0.9.1 API Endpoints

```
# Segmentation (AI)
POST /api/segment/point      # Point-based prompting (SAM-Med3D)
POST /api/segment/box       # Bounding box prompting
POST /api/segment/refine    # Refine with negative points

# Volume Management
POST /api/volumes/upload    # Upload DICOM / NIfTI
GET  /api/volumes/{id}     # Retrieve processed volume

# Mesh Generation
POST /api/mesh/from-mask    # Mask → MZ3 mesh (nii2mesh)
POST /api/mesh/quick        # Fast preview (marching cubes)

# MEDIS TXT Processing
POST /api/medis/upload      # Upload MEDIS TXT
POST /api/medis/to-mesh     # Convert contours to mesh

# Straightened MPR
POST /api/straighten/create  # Centerline → straightened volume

# Batch Processing
POST /api/batch/process     # Queue batch of cases
GET  /api/batch/{job_id}    # Job status

# Authentication
POST /api/auth/login        # LDAP / Charité SSO
GET  /api/health            # Health check + GPU status
```

0.9.2 HU Intensity Normalisation (CCTA Pre-processing)

SAM-Med3D was trained on data with modality-specific intensity normalisation. Raw CCTA HU values span -1000 to $+3000$, but only the cardiovascular window is relevant. Feeding unnormalised values will degrade segmentation quality.

Required pre-processing before inference:

```
def normalise_ccta(volume: np.ndarray, clip_min: float = -100, clip_max: float = 700)
    """
    Clip to cardiovascular window and normalise to [0, 1].
```

```
clip_min=-100 HU: excludes air; clip_max=700 HU: captures calcification peak.
Adjust clip_max to 400 HU for lumen-only work (excludes dense calcium blooming)
"""
volume = np.clip(volume, clip_min, clip_max)
volume = (volume - clip_min) / (clip_max - clip_min) # → [0, 1]
return volume.astype(np.float32)
```

Note: The exact normalisation used in SAM-Med3D's SA-Med3D-140K training pipeline is not fully documented for CT modalities. The values above are a clinically reasonable starting point. Measure Dice vs. normalisation window on a held-out validation set and tune accordingly. For prostate mpMRI, normalise each sequence (T2W, ADC) independently using per-volume z-score normalisation.

0.9.3 Segmentation Endpoint (Core)

```
from typing import Optional
from fastapi import APIRouter, HTTPException
from pydantic import BaseModel
import nibabel as nib
import numpy as np
import torch
import medim

router = APIRouter()

class SegmentRequest(BaseModel):
    volume_id: str
    coordinates: list[list[float]] # [[x,y,z], ...] in mm – one or more points
    labels: Optional[list[int]] = None # 1=positive, 0=negative per point; default

# Model loaded once at startup (inside FastAPI lifespan event in production)
model = medim.create_model(
    "SAM-Med3D", pretrained=True,
    checkpoint_path="app/models/checkpoints/sam_med3d_turbo.pth"
)
device = "cuda" if torch.cuda.is_available() else "cpu"
model = model.to(device)
if device == "cuda":
    model = model.half()
model.eval()

# Use def (not async def): FastAPI runs sync endpoints in a thread pool,
# keeping the event loop free during blocking GPU inference.
@router.post("/api/segment/point")
```

```

def segment_point(req: SegmentRequest):
    if not req.coordinates:
        raise HTTPException(status_code=400, detail="At least one coordinate is req
    if any(len(pt) != 3 for pt in req.coordinates):
        raise HTTPException(status_code=400, detail="Each coordinate must be [x, y,
    if req.labels is not None and len(req.labels) != len(req.coordinates):
        raise HTTPException(status_code=400, detail="labels length must match coord

    # Load volume from cache/disk
    vol_nii = nib.load(f"/data/volumes/{req.volume_id}.nii.gz")
    volume = vol_nii.get_fdata(dtype=np.float32)

    # Normalise HU values to cardiovascular window before inference
    volume = normalise_ccta(volume)

    # Convert mm → voxel coordinates for each point
    inv_affine = np.linalg.inv(vol_nii.affine)
    voxel_points = [(inv_affine @ [*pt, 1])[:3] for pt in req.coordinates]

    effective_labels = req.labels if req.labels is not None else [1] * len(voxel_po

    # Run SAM-Med3D inference – no_grad disables gradient tracking to halve VRAM us
    with torch.no_grad():
        mask = model.segment(volume, points=voxel_points, labels=effective_labels)

    # Return mask as NIfTI
    mask_nii = nib.Nifti1Image(mask.astype(np.uint8), vol_nii.affine)
    # ... serialize and return

```

0.10 MEDIS TXT Reference Pipeline

0.10.1 Format Description

MEDIS TXT contains vessel wall contours from MEDIS QAngio CT: - **Lumen** contours: define inner wall (blood pool boundary) - **VesselWall** contours: define outer wall (including plaque) - Each contour: group label, slice distance, N points in 3D mm coordinates

0.10.2 Parser (TypeScript)

```

export interface MedisContour {
    group: "Lumen" | "VesselWall";
    sliceDistance: number;           // mm along vessel
    points: { x: number; y: number; z: number }[];

```



```

}

export function parseMedisTxt(content: string): MedisContour[] {
  const lines = content.split("\n");
  const contours: MedisContour[] = [];
  let current: Partial<MedisContour> = {};
  let points: { x: number; y: number; z: number }[] = [];

  for (const line of lines) {
    if (line.startsWith("# group:")) {
      current.group = line.split(":")[1].trim() as "Lumen" | "VesselWall";
    } else if (line.startsWith("# SliceDistance:")) {
      current.sliceDistance = parseFloat(line.split(":")[1]);
    } else if (line.startsWith("# Contour index:")) {
      if (current.group && points.length > 0) {
        contours.push({ ...current, points } as MedisContour);
      }
      // Reset both points AND current so the next contour does not inherit
      // stale group/sliceDistance values from the previous block.
      current = {};
      points = [];
    } else if (line.trim() && !line.startsWith("#")) {
      const [x, y, z] = line.trim().split(/\s+/).map(Number);
      if (!isNaN(x)) points.push({ x, y, z });
    }
  }
  if (current.group && points.length > 0) {
    contours.push({ ...current, points } as MedisContour);
  }
  return contours;
}

```

0.10.3 Direct Client-Side Mesh Construction (< 100 ms)

Contour rings are connected into a tube mesh directly in the browser — no backend round-trip needed. Algorithm: connect ring N to ring N+1 with triangle pairs.

Performance comparison: | Method | Latency | Network | | Backend (build-stl.py → STL → download) | 500–2000 ms | Required | | **Client-side (TXT → NVMesh) | 50–100 ms | None |**

0.11 Mesh Generation Strategies

0.11.1 Three Approaches

Approach	Method	Speed	Quality	Use Case
Ultra-Simple	Connect contour rings → STL	< 50 ms	Faceted	MEDIS export
Client-side vtk.js	Marching cubes on mask	< 1 s	Good	Interactive preview
Server nii2mesh → MZ3	Decimation + smoothing	1–3 s	High	Final visualisation

0.11.2 Recommended Web Format: MZ3

- 3–5× smaller than PLY, 10× smaller than STL
- Binary, gzip-compressed, native Niivue support
- Target: < 5 MB per mesh, 50K–200K triangles, 60 FPS rendering

0.12 Centerline Extraction & Straightened MPR (CPR)

0.12.1 Overview

Straightened MPR (Curved Planar Reformation) “unfolds” a tortuous vessel into a straight view. Essential for assessing stenosis and plaque distribution along the entire vessel length.

Three steps: 1. **Centerline extraction** — centroid of lumen contours (from MEDIS) or Voronoi skeletonisation (from AI mask) 2. **Bishop (parallel transport) frame** — compute Tangent (T) then propagate Normal (N) and Binormal (B) without relying on curvature (see Mathematical Foundation for why Frenet-Serret N must not be used) 3. **Cross-section sampling** — extract perpendicular slices → stack into straightened volume

0.12.2 Mathematical Foundation

Tangent (finite differences):

```
T[i] = normalize(centerline[i+1] - centerline[i-1]) // central difference
```

Normal — Bishop (parallel transport) frame:

Why not Frenet-Serret: The Frenet-Serret principal normal $N = \text{normalize}(dT/ds)$ is undefined and flips 180° in low-curvature (near-straight) vessel segments, producing rotating or jumping cross-sections. Bishop frame (rotation-minimizing frame) propagation^{9,10} avoids this by carrying an initial normal forward without relying on curvature.

```
// Initialise with any vector perpendicular to T[0]
```

```
N[0] = arbitrary_perpendicular(T[0])
```

```
// Propagate: project previous N onto the plane perpendicular to new T
for i in 1..n-1:
```

$$N[i] = \text{normalize}(N[i-1] - \text{dot}(N[i-1], T[i]) * T[i])$$

Binormal: $B[i] = T[i] \times N[i]$

Cross-section point: $Q(u,v) = P[i] + u \cdot N[i] + v \cdot B[i]$

0.12.3 Interactive Controls

- **Position slider:** Select centerline point ($0 \rightarrow N-1$)
- **Rotation slider:** Rotate cross-section around T (Rodrigues' formula)
- **Zoom slider:** Adjust cross-section FOV
- **Quad-view:** CCTA overview | cross-section | straightened MPR | 3D mesh

0.13 Deployment, Performance & Security

0.13.1 Hardware Requirements

0.13.1.1 Development Environment

Component	Minimum	Recommended	Current Setup
GPU	RTX 3060 (8 GB)	RTX 2080 Ti (11 GB)	2× RTX 2080 Ti (11 GB each)
CPU	6-core	8+ core	Modern workstation
RAM	16 GB	32 GB+	Sufficient
Storage	500 GB SSD	1 TB+ NVMe	Adequate
CUDA	11.8+	12.2+	CUDA 12.2

Development use cases: model testing, single-user interactive segmentation, DISCHARGE sub-set processing, active-learning experiments.

0.13.1.2 Production Environment (Charité Clinical Deployment)

Component	Minimum	Recommended	Notes
GPU Cluster	2× RTX 4090 (24 GB)	4× A100/H100 (40–80 GB)	Concurrent clinical users
CPU	16-core	32+ core	FastAPI + preprocessing
RAM	64 GB	128 GB+	Multiple 3D volumes in memory
Storage	10 TB+	50 TB+	DISCHARGE + clinical data
Network	10 Gbps	25 Gbps+	Fast volume transfers

Component	Minimum	Recommended	Notes
Redundancy	RAID 10	Distributed storage	Clinical data safety

Production requirements: GDPR compliance (all data on-premise), 99.9 % uptime, 5–10 concurrent radiologists, full DISCHARGE batch processing, automated backup and failover.

0.13.2 GDPR & Deployment Constraints

WARNING: IN-HOUSE BACKEND MANDATORY FOR CLINICAL USE

For any clinical deployment, the SAM-Med3D-turbo model **MUST** run on Charité’s on-premise infrastructure:

Requirement	Reason	Alternative
On-premise GPU servers	GDPR — patient data cannot leave hospital	Not allowed: cloud inference
Charité network	Secure medical data transmission	Not allowed: public internet
Hospital authentication	User access control and audit trails	Not allowed: anonymous access
Medical device certification	Clinical safety and regulatory compliance	Not allowed: research-only deployment

Browser (Hospital Network) → Charité Firewall → Internal GPU Cluster

↓	↓	↓
UI/UX + Niivue	Load Balancer	SAM-Med3D-turbo
3D visualization	+ Authentication	PyTorch Inference
User prompts	+ Logging	+ Medical Data Store

Non-clinical use cases (anonymised data allowed): research demos with cloud GPU; local development with sample datasets; open-source contributions via GitHub with synthetic data; educational browser-only UI with mock backend.

0.13.3 Infrastructure

Component	Technology
Containerisation	Docker + NVIDIA Container Toolkit
GPU	4× NVIDIA A100 (80 GB each)
Embedding cache	Redis (sub-second for repeated prompts)
Batch queue	Celery + Redis broker
Authentication	LDAP / Charité SSO
Compliance	GDPR (all data on-premise, no cloud)

0.13.4 Performance Targets

Metric	Target
Single-click → mask	< 2 s end-to-end
Embedding computation (first prompt)	~1 s
Subsequent prompts (cached embedding)	< 0.5 s
Mesh generation (MZ3)	< 3 s
Batch throughput	~50 cases / hour (4 GPUs) — bottleneck is DICOM→NIfTI preprocessing, not GPU
CTA volume memory	~268 MB ($512^3 \times 2$ bytes = 268,435,456 bytes)

0.13.5 Memory Budget

Component	Size
CTA volume (512^3 , int16)	~268 MB
SAM-Med3D-turbo (FP16)	~4 GB VRAM
Embedding cache (per volume)	~50–500 MB (estimate — depends on patch count and feature map size; to be measured)
Straightened volume ($64^2 \times 200$)	~8 MB
Mesh (MZ3, per vessel)	< 5 MB

0.14 Research Roadmap & Milestones

0.14.1 Phase 1: MVP — MEDIS TXT Visualisation (Weeks 1–4)

- ☐ MEDIS TXT parser + client-side mesh in Niivue
- ☐ Centerline extraction + straightened MPR with **Bishop (parallel transport) frame** (not Frenet-Serret — see Mathematical Foundation)
- ☐ Quad-view layout with interactive sliders
- ☐ NIfTI.gz / DICOM loading

0.14.2 Phase 2: SAM-Med3D Integration (Weeks 5–8)

- ☐ Backend: load turbo checkpoint, expose `/api/segment/point`
- ☐ Frontend: click → prompt → mask overlay
- ☐ Redis embedding cache for sub-second repeated prompts
- ☐ **Implement dense_prompt support** — either (a) custom prompt encoder modification to accept a prior mask, or (b) lumen mask surface → additional positive point prompts; required blocker for dual-wall pipeline
- ☐ Dual-wall sequential segmentation pipeline (depends on above)

0.14.3 Phase 3: DISCHARGE Evaluation (Weeks 9–12)

- ☐ **Zero-shot baseline — myocardium first:** LV_MY0 is the closest match to ACDC pre-training targets; establish Dice baseline here before coronary lumen (see Verified Performance)
- ☐ Zero-shot baseline on coronary lumen (held-out DISCHARGE cases)
- ☐ Quantify Dice, Hausdorff, diameter stenosis % correlation vs. MEDIS QAngio CT expert measurements
- ☐ Identify failure modes (motion, calcification, bifurcations, patch-boundary seams)

0.14.4 Phase 4: Fine-tuning & Active Learning (Weeks 13–20)

- ☐ Fine-tune SAM-Med3D on DISCHARGE annotations (nnU-Net-style data prep)
- ☐ Active-learning loop¹¹: expert corrections → weekly re-training. **Correction data model:** a “correction” is a full per-structure binary NIfTI mask saved by the radiologist after local brush edits in the browser. The backend must convert the in-browser voxel edits (sparse difference from AI mask) into a complete corrected NIfTI matching Data Format for Fine-tuning before ingestion into the fine-tuning pipeline.
- ☐ Benchmark against task-specific nnU-Net baseline

0.14.5 Phase 5: Prostate Extension (Weeks 21–28)

- ☐ Adapt pipeline for prostate mpMRI (multi-sequence input)
- ☐ Zone-specific class labels (PZ / TZ / lesion)
- ☐ Validate on Charité prostate cohort

0.14.6 Phase 6: Clinical Validation & Publication (Weeks 29–36)

- ☐ Prospective reader study (Dice vs. time vs. inter-observer)
- ☐ Open-source release + MedSegFM competition baseline
- ☐ Publication: “Foundation-model-assisted coronary CCTA segmentation at scale”

0.14.7 Quarterly Research Milestones

Quarter	Milestone	Status
Q1 2026 (ends 2026-03-31)	Zero-shot baseline on DISCHARGE + MVP deployed	In progress - update before end of quarter
Q2 2026	Active-learning loop running; Dice $\geq$ 0.80 on coronary lumen	Pending
Q3 2026	Prospective reader study; prostate pipeline validated	Pending
Q4 2026	Open-source release; conference/journal submission	Pending

0.15 Related Medical Segmentation Models

Model	Dimension	Modalities	Prompts	Key Difference from SAM-Med3D
SAM (Meta)	2D	Natural images	Point/box/text	No medical training, 2D only
SAM-Med2D	2D	Medical (slice-wise)	Point/box	2D → cannot capture volumetric context
MedSAM	2D	Medical (slice-wise)	Box only	Simpler architecture, box prompts only
SAM-Med3D	3D	Medical (volumetric)	3D point	Native 3D — our choice
nnU-Net	3D	Medical (task-specific)	None (automatic)	Not promptable; requires per-task training

This document is the living foundation of the Charité Segment Platform research project. All claims about SAM-Med3D are verified against the published paper (arXiv:2310.15161), official GitHub README, and HuggingFace model/dataset cards. Critical caveats about zero-shot performance on coronary CTA and prostate mpMRI (neither directly evaluated in the paper) are noted throughout. Full bibliography in `plan/references.bib`.

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